Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 13 complete test ICs for all5 lambdas within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**0**	0.0381	0.115	1.07	1.02	0.329	1	0.241	1.02	1.01	0	18/18
1 ANN10	**1**	0.235	0.873	5.92	0.976	0.613	1.55	0.344	3.92	6.24	0	17/18
1 ANN10	**10**	0.323	0.812	6.86	8.07	0.791	6.81	0.329	163	397	0	14/18
1 ANN10	**100**	0.658	0.669	1	1	1	31.4	1	1.03	34.6	0	18/18
1 ANN10	**1000**	0.66	1.29	1	89.1	1.59	5.29	1	1	756	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 4, 5, 7, 18, 39, 44, 53, 64, 68, 69, 73, 78, 88. IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 6 complete test ICs for all5 lambdas within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
2 P-only DeepONet10	**0**	0.0463	0.0234	0.782	0.86	0.118	0.844	0.0596	0.609	0.683	0	18/18
2 P-only DeepONet10	**1**	0.0978	0.199	0.964	2.72	0.813	3.66	0.185	4.86	2.84	0	18/18
2 P-only DeepONet10	**10**	0.564	0.902	1.06	1	5.65	3.15	1	253	8.52e+03	0	18/18
2 P-only DeepONet10	**100**	0.606	0.902	1	1	8.05	14.4	1	1.01	1.05	0	6/18
2 P-only DeepONet10	**1000**	0.571	1.04	1	1	5.69	1	1	2.53e+04	1	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 4, 18, 44, 53, 73, 78. IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 18 complete test ICs for all5 lambdas within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
3 ANN10 + P-only DeepONet10	**0**	0.0373	0.0777	0.901	0.92	0.244	0.993	0.0941	0.995	1	0	18/18
3 ANN10 + P-only DeepONet10	**1**	0.0437	0.454	0.69	5.28	0.594	0.985	0.0777	2.95	1.44	0	18/18
3 ANN10 + P-only DeepONet10	**10**	0.322	0.708	4.98	9.89	0.756	7.17	0.559	629	5.9e+03	0	18/18
3 ANN10 + P-only DeepONet10	**100**	0.647	0.928	1	1	2	31.2	1	1.17	38.7	0	18/18
3 ANN10 + P-only DeepONet10	**1000**	0.542	1.64	1	1	5	1	1	0.998	815	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 8 complete test ICs for all5 lambdas within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
4 T/P DeepONet10	**0**	0.0822	0.361	1.2	0.731	0.368	0.924	0.231	0.993	0.995	0	17/18
4 T/P DeepONet10	**1**	0.293	0.749	3.78	1.25	0.58	1.12	0.496	14	22.7	0	9/18
4 T/P DeepONet10	**10**	0.565	0.922	7.04	2.49	1.03	1.97	0.832	32.5	1	0	15/18
4 T/P DeepONet10	**100**	0.699	0.8	2.95	3.54	0.705	23.9	1	1.02	1	0	18/18
4 T/P DeepONet10	**1000**	0.548	1.42	1	1	0.988	1	1	2.47e+03	1	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 4, 5, 39, 44, 53, 64, 68, 69. IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 18 complete test ICs for all5 lambdas within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
5 ANN10 + T/P DeepONet10	**0**	0.0491	0.0957	0.962	0.82	0.174	0.97	0.066	0.921	0.993	0	18/18
5 ANN10 + T/P DeepONet10	**1**	0.0803	0.41	0.932	0.819	0.535	0.822	0.103	0.938	1.16	0	18/18
5 ANN10 + T/P DeepONet10	**10**	0.216	0.948	2.49	6.19	0.814	3.29	0.323	55.4	129	0	18/18
5 ANN10 + T/P DeepONet10	**100**	0.332	0.973	1.42	11.5	2.46	1	0.561	324	2.03e+03	0	18/18
5 ANN10 + T/P DeepONet10	**1000**	0.542	1.64	1	1	5	1	1	0.998	761	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 8 complete test ICs for all5 lambdas within T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
6 T/P/ϕ DeepONet10	**0**	0.141	0.293	1	1.02	0.425	1.4	1	1.2	1.46	0	17/18
6 T/P/ϕ DeepONet10	**1**	0.651	0.966	16.9	71.6	0.674	7.13	0.855	14	1	0	11/18
6 T/P/ϕ DeepONet10	**10**	0.489	0.806	11.1	5.98	0.735	3.29	1.01	9.32	2.4	0	17/18
6 T/P/ϕ DeepONet10	**100**	0.708	0.613	26	34	0.684	18.9	0.926	1	1	0	16/18
6 T/P/ϕ DeepONet10	**1000**	0.533	1.4	1	1	1.36	1	1	6.21	1	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 5, 7, 39, 44, 53, 68, 69, 78. IC sets differ between teams; use the all35-variant summary for architecture comparisons.

Lambda sweep | Cantera vs NN NRMSE summary

Median of per-IC NRMSE on the same 17 complete test ICs for all5 lambdas within ANN10 + T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	lambda	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
7 ANN10 + T/P/ϕ DeepONet10	**0**	0.0672	0.121	1	0.606	0.21	1	0.245	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1**	0.395	0.881	2.89	3.51	0.521	8.19	0.507	1.22	1.15	0	17/18
7 ANN10 + T/P/ϕ DeepONet10	**10**	0.159	0.457	2.44	5.75	0.832	3.11	0.196	68	45.7	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**100**	0.349	0.865	1.55	11.3	3.53	1	0.587	358	2.88e+03	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**1000**	0.545	1.47	1	1	5.06	1	1	0.998	752	0	18/18

Exact user util.py defaults: norm="rms", reference=Cantera; equal time-sample weights, dimensionless ratio without x100.
N/A: undefined zero-reference RMS or empty common set. —: early stop, so no full-horizon NRMSE. Nonzero constant N2 is valid.
Every network has10000 updates. Lambda0/10 controls are reused; lambda1/100/1000 are fresh paired-seed models.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 64, 68, 69, 73, 74, 76, 78, 88. IC sets differ between teams; use the all35-variant summary for architecture comparisons.